# Executive Summary

*Bayesian Logistic Regression and Hierarchical Modeling for Alzheimer's Disease Prediction*

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## 1. The Problem

Alzheimer's disease is a leading cause of dementia worldwide. **Early identification** of at-risk patients enables preventive strategies that may delay cognitive decline but traditional logistic regression only provides **point estimates** (e.g., "risk = 0.7") without telling clinicians **how certain** that prediction is.

In a clinical setting, this matters. A prediction of 0.7 with wide uncertainty is fundamentally different from a prediction of 0.7 with tight uncertainty, yet standard models cannot express this distinction.

**The gap:** Clinicians need predictions with **credible intervals**, not just point estimates.

## 2. Approach

Using demographic, lifestyle, medical history, and cognitive assessment data from **2,149 older adults** (ages 60–90), we built two Bayesian models in **Stan**:

1. **Baseline Bayesian Logistic Regression** — probabilistic classification with weakly informative priors.
2. **Hierarchical Logistic Regression** — varying intercepts by ethnicity group (Caucasian, African American, Asian, Other) using partial pooling.

Continuous predictors (MMSE, ADL, BMI) were standardized to improve sampler geometry. Priors were centered on the empirical AD prevalence (0.354) with weakly informative scales to regularize effects and mitigate overfitting.

## 3. Key Results at a Glance

| Finding | What It Means |
|---|---|
| **Bayesian inference works** | All parameters converged ($\hat{R} \approx 1.00$); posterior predictive checks confirmed calibration. |
| **Cognitive measures dominate** | Memory Complaints (OR: 13.4) and Behavioral Problems (OR: 12.0) are strongest risk factors. |
| **Hierarchical adds little** | $\sigma_\alpha = 0.41$ captures moderate ethnicity variation but does not improve prediction. |
| **Strong discrimination** | AUC $\approx 0.86$ for both models; baseline preferred for efficiency. |

## 4. Key Findings in Detail

### 1. Cognitive measures are the strongest predictors

Posterior distributions identified **Memory Complaints** (OR: 13.4) and **Behavioral Problems** (OR: 12.0) as the strongest risk factors. Protective effects were seen for **MMSE**, **ADL**, and **Functional Assessment** scores—all with 95% credible intervals excluding zero.

> **Key Finding:** Clinically, this confirms that cognitive and functional assessments carry more signal than static demographic factors.

### 2. Hierarchical modeling captures ethnicity variation but does not improve prediction

The hierarchical model estimated $\sigma_\alpha = 0.41$, indicating **moderate between-ethnicity variation** in baseline risk. However, LOO-CV and WAIC showed **no predictive improvement** over the simpler baseline model.

> **Key Finding:** The simpler model achieves equivalent accuracy with greater computational efficiency; hierarchical modeling remains valuable for subgroup analysis and equity research.

### 3. Bayesian inference enables full uncertainty quantification

Every prediction comes with a **full posterior distribution** and **95% credible intervals**—not just a point estimate. Posterior predictive checks confirmed the models reproduce the observed outcome distribution.

> **Key Finding:** Clinicians can ask "what is the probability this patient's risk exceeds 0.5?"—a question frequentist models cannot answer directly.

## 5. Key Predictors with 95% Credible Intervals

| Predictor | Log-Odds | 95% CI | Interpretation |
|---|---|---|---|
| **Memory Complaints** | 2.60 | [2.38, 2.81] | Strongest risk factor |
| **Behavioral Problems** | 2.49 | [2.26, 2.72] | Strong risk factor |
| **Functional Assessment** | −0.45 | [−0.49, −0.42] | Protective |
| **Activities of Daily Living (ADL)** | −0.42 | [−0.46, −0.39] | Protective |
| **Mini-Mental State Exam (MMSE)** | −0.11 | [−0.12, −0.10] | Protective |

## 6. Practical Implications

- **Uncertainty-aware risk stratification**: Clinicians receive probability scores with credible intervals, supporting shared decision-making.
- **Model selection is a clinical choice**: For prediction, the baseline model is preferred; for equity research, the hierarchical model reveals subgroup differences.
- **Regularization prevents overfitting**: Weakly informative priors stabilize estimates in high-dimensional clinical data.

## 7. Skills Demonstrated

| Skill Area | Specific Techniques |
|---|---|
| **Bayesian Inference** | Stan, Hamiltonian Monte Carlo, MCMC diagnostics |
| **Hierarchical Modeling** | Partial pooling, varying intercepts, non-centered parameterization |
| **Model Comparison** | LOO-CV, WAIC, posterior predictive checks |
| **Uncertainty Quantification** | 95% credible intervals, full posterior distributions |
| **Statistical Computing** | R, `rstan`, `bayesplot`, `loo`, `dplyr`, `ggplot2` |
| **Clinical Application** | Alzheimer's risk stratification, health equity analysis |

## 8. Limitations

- Specificity exceeded sensitivity—suggesting the model is better at ruling out AD than detecting early disease.
- Cross-sectional data limits causal interpretation.
- External validation on independent datasets is needed to confirm generalizability.
- Ethnicity was treated as a group-level intercept; heterogeneous predictor effects (varying slopes) were not modeled.

## 9. Main Conclusion

Bayesian inference provides a rigorous foundation for Alzheimer's risk prediction with full uncertainty quantification. While the hierarchical model captures moderate ethnicity-level variation, the **baseline Bayesian logistic regression offers the best balance of accuracy and computational efficiency**. Cognitive measures—particularly Memory Complaints and Behavioral Problems—emerge as the strongest predictors, reinforcing the clinical value of cognitive assessment in early AD detection.

**Full Paper and Reproducible Code (Stan):**
github.com/your-username/bayesian-alzheimers-prediction

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**Keywords:** Bayesian inference · Hierarchical modeling · Alzheimer's disease · MCMC · Stan · LOO-CV · WAIC · Uncertainty quantification